Eigenvalues and Eigenvectors

Lecturer: Chng Zhi Yee (zy_chng@um.edu.my)
Based on the lecture note uploaded on SPeCTRUM

Sem I, 2025/2026

Learning Outcomes

At the end of this lecture, you should be able to:

☐ Know what is eigenvalues and eigenvectors of a matrix.

☐ Determine characteristic polynomial and characteristic equation of a matrix.

☐ Find the eigenvalues of a matrix and its corresponding eigenvector.

Let $A \in M_n(\mathbb{R})$. A scalar λ is called an eigenvalue of A if there exists a nonzero column vector $X \in M_{n,1}(\mathbb{R})$ such that $AX = \lambda X$. Such X is called an eigenvector of A corresponding to (or associated with) the eigenvalue λ.

Example:

Let $A = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix}$, $X_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$.

$AX_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} = -1X_1$ | $AX_2 = \begin{bmatrix} 5 \\ 10 \end{bmatrix} = 5X_2$

Eigenvalues and eigenvectors

Example:

Let $A = \begin{bmatrix} 1 & -2 \\ -1 & 0 \end{bmatrix}$, $X_1 = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $X_3 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$.

$\begin{bmatrix} 4 \\ -2 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$ $\qquad$ $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$2X_1$ $\qquad$ $-X_2$

Example:
Let $A \in M_n(\mathbb{R})$. Show that

(i) If $\lambda \in \mathbb{R}$ is an eigenvalue of A, then λ^n is an eigenvalue of A^n for every positive integer n.

(ii) If $X \in M_{n,1}(\mathbb{R})$ is an eigenvector of A corresponding to the eigenvalue $\lambda \in \mathbb{R}$, then, for every nonzero scalar $\alpha \in \mathbb{R}$, the vector αX is an eigenvector of A corresponds to the eigenvalue λ.

$AX = \lambda X$

$A^n X$

$= A^{n-1}[AX]$

$= A^{n-1}[\lambda X]$

$\vdots$

$= \lambda^n X \quad \square$

$Ax = \lambda x$

$\alpha A x = \alpha \lambda x$

$A(\alpha x) = \lambda(\alpha x)$, Let $X = \alpha x$

$AX = \lambda X \quad \square$

Let $A \in M_n(\mathbb{R})$. A scalar λ is an eigenvalue of A if and only if $\det(A - \lambda I_n) = 0$.

$Ax = \lambda x$

$Ax - \lambda x = 0$

$|(A - \lambda I)(x)| = |0|$

$Bx = 0$

$|A - \lambda I||x| = 0$

since $|x| \neq 0$, [$x=0$ is trival sol]

$|A - \lambda I_n| = 0$

Let $A \in M_n(\mathbb{R})$. Then, $p(\lambda) = \det(A - \lambda I_n)$ is called a characteristic polynomial of A and $p(\lambda) = \det(A - \lambda I_n) = 0$ is the characteristic equation of A.

Finding eigenvalues and eigenvectors

Procedure to determine eigenvalues and eigenvectors of A

1. For a given $A \in M_n(\mathbb{R})$, determine the characteristic polynomial.

$$p(\lambda) = \det(A - \lambda I_n) = \begin{vmatrix} a_{11} - \lambda & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} - \lambda & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{11} \cancel{- \lambda} & a_{12} & \cdots & a_{1n} - \lambda \end{vmatrix}.$$

2. Solve the characteristic equation $p(\lambda) = 0$ and determine all the eigenvalues λ_i.

3. For each λ_i, solve for v_i in equation $Av_i = \lambda_i v_i$, or equivalently, $(A - \lambda_i I_n)v_i = 0$, and find an eigenvector v_i corresponding to λ_i.

Finding eigenvalues and eigenvectors

Example:

Determine the characteristic equation of $A = \begin{bmatrix} 1 & 1 & -2 \\ -1 & 2 & 1 \\ 0 & 1 & -1 \end{bmatrix}$. Then for each λ_i, find an eigenvector v_i corresponding to λ_i.

$$\lambda = \{-1, 1, 2\}$$

$$\begin{pmatrix} t \\ 0 \\ t \end{pmatrix} \quad \begin{pmatrix} 3t \\ 2t \\ t \end{pmatrix} \quad \begin{pmatrix} t \\ 3t \\ t \end{pmatrix}$$

Finding eigenvalues and eigenvectors

Finding eigenvalues and eigenvectors

Finding eigenvalues and eigenvectors

Example:

Determine all the eigenvalues and eigenvectors of $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$.

$$\lambda = \cos\theta + i\sin\theta$$

$$\left\{ \begin{pmatrix} \frac{-i\sin\theta}{\sin\theta} t \\ t \end{pmatrix}, \ t \neq 0, \ \sin\theta \neq 0 \right\}$$

Finding eigenvalues and eigenvectors

Finding eigenvalues and eigenvectors

Example:

Determine the characteristic equation of $A = \begin{bmatrix} 2 & 2 & -1 \\ -6 & -6 & 3 \\ -4 & -4 & 2 \end{bmatrix}$. Then for each λ_i, find an eigenvector v_i corresponding to λ_i.

$\lambda = -2, 0, 0$

$\begin{pmatrix} -\frac{1}{2}t \\ \frac{3}{2}t \\ t \end{pmatrix}$ $\begin{pmatrix} \frac{t-2h}{2} \\ h \\ t \end{pmatrix}$

$\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ $\begin{pmatrix} -\frac{1}{2} \\ 1 \\ 1 \end{pmatrix}$ $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

Finding eigenvalues and eigenvectors

Some ways to check your work

$[\ \diagdown\ ] = \Sigma \lambda$

- The sum of the eigenvalues equals the trace of the matrix (the sum of the diagonal). When adding up the eigenvalues, each eigenvalues should be added up to its multiplicity.
- When finding the eigenvectors, there must be always an infinitely number of solutions. $|A - \lambda I_n| = 0$
- Once you have found λ and v, check that $Av = \lambda v$.

$$\sum_{j=1}^{n} \sum_{i=1}^{n} (a_{ii} - \lambda_j) = 0$$

$$n \sum_i a - n \sum_j \lambda = 0$$

$$\Sigma a = \Sigma \lambda$$